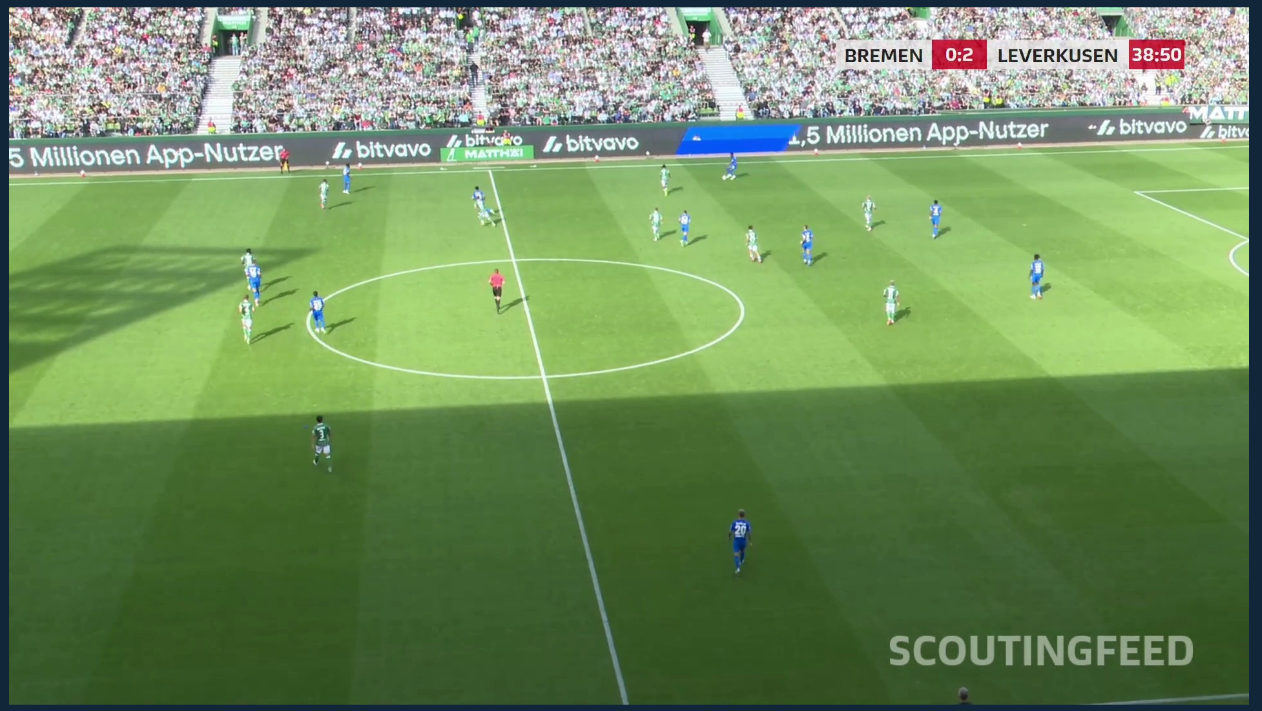
Preprocessing effects on fixed pose models

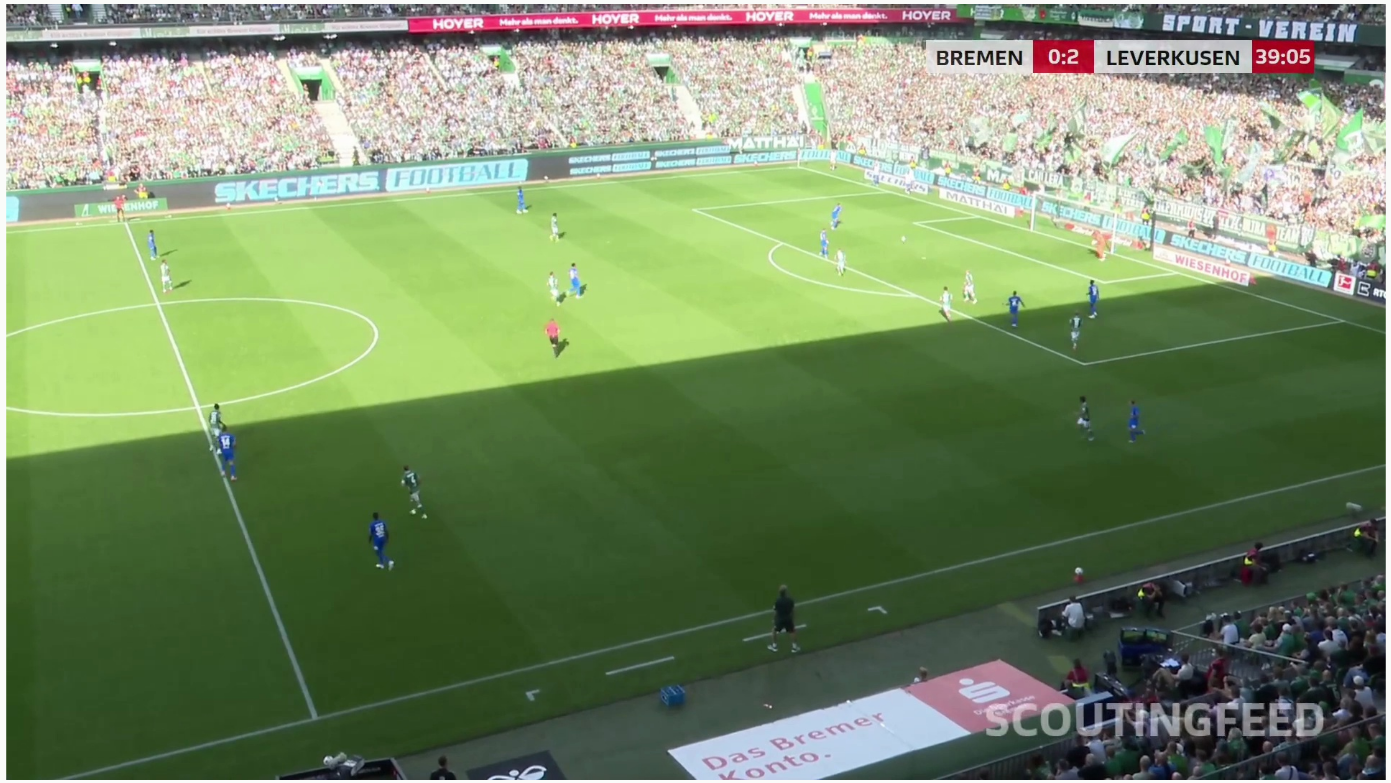
Controlled experiments in football video



Master’s thesis
Scientific design and current status

Research question

How does preprocessing change pose quality when model weights stay fixed?



Independent variable

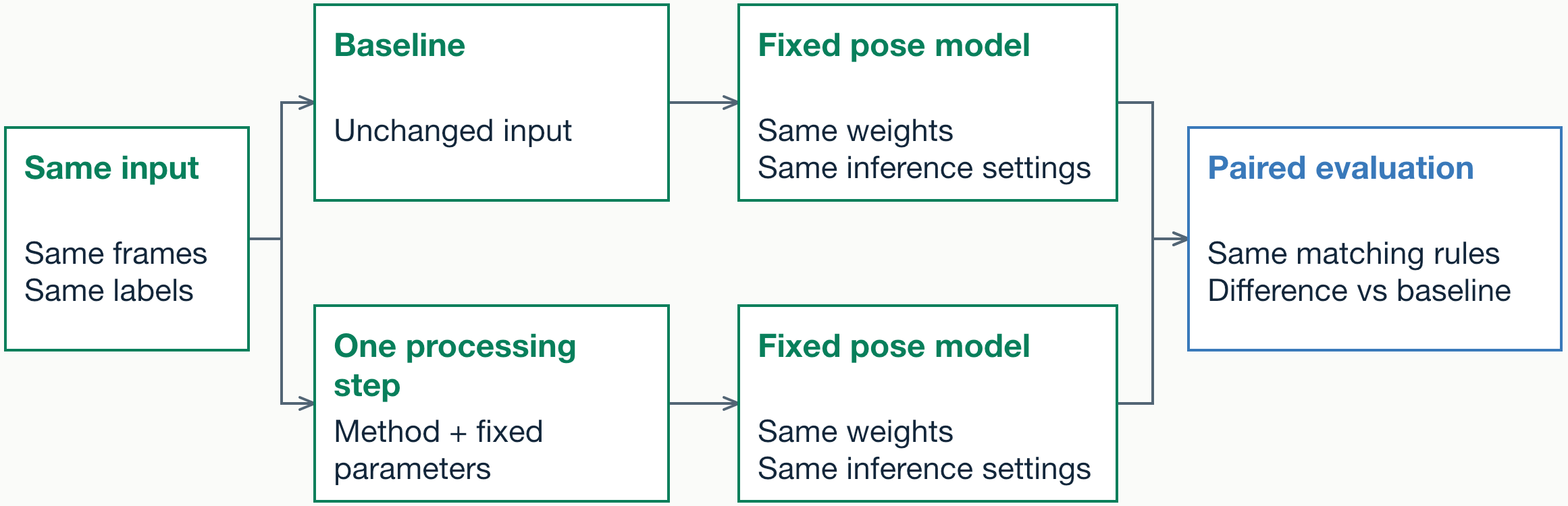
Processing method and its parameters

Measured outcomes

Detection coverage
Joint localization
Computational cost

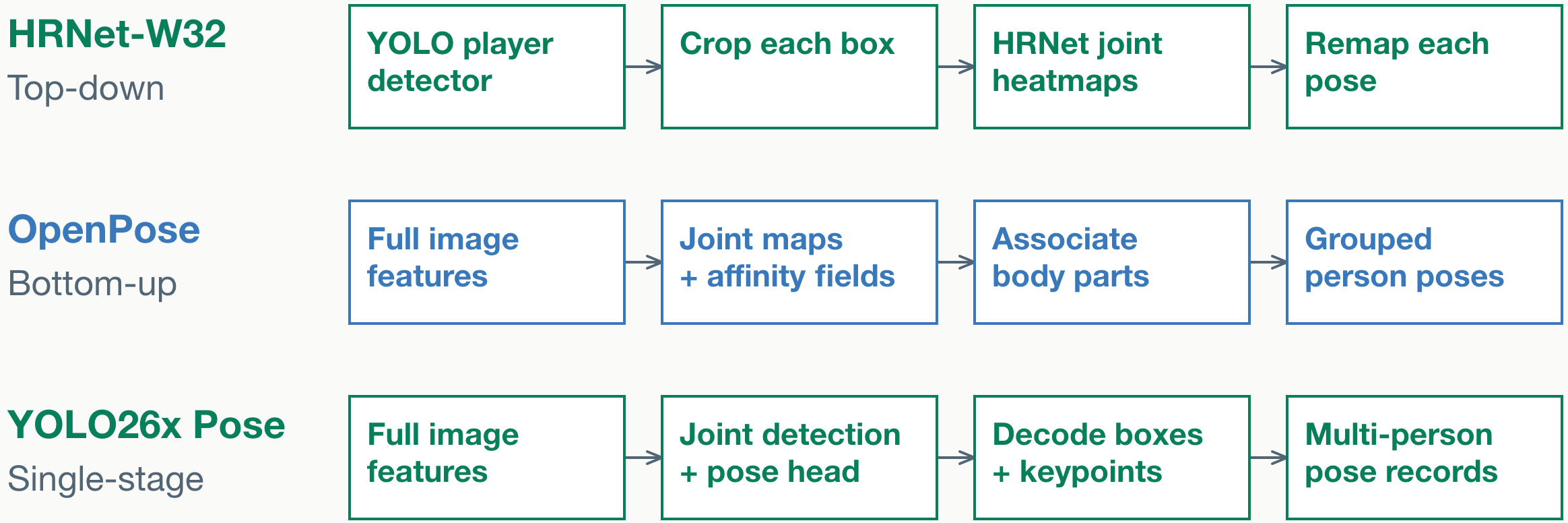
Controlled comparison

Fixed published pretrained checkpoints. No training or fine-tuning within the comparison.



Three routes to multi-person poses

“Multi-person” is an output capability shared by all three, not a separate architecture.



Ground truth for distinct evaluation tasks

Reference data must describe the quantity that each metric evaluates.

Evaluation task	Independent reference	Current position
COCO pose benchmark	Official person keypoints and visibility labels	Available
Football image-plane pose	Validated DFL/Werder joints, if available Otherwise reviewed manual annotations	Format and alignment still pending
Pitch-coordinate positions	Synchronized tracking x/y with independently checked calibration	Later evaluation stage

Football reference-data protocol

Proposed workflow before the final accuracy experiment

1. **Define the evaluation sample**

   Include varied scale, lighting, occlusion and camera views

2. **Align the reference to the video**

   Match timestamps, player identities and joint definitions

3. **Review labels independently**

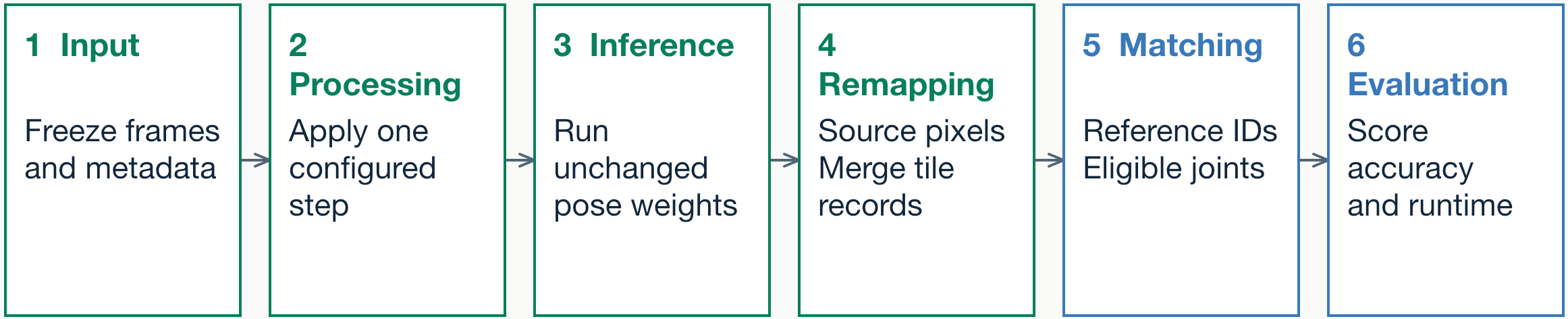
   Record visibility, uncertainty and annotation disagreements

4. **Freeze development and test sets**

   Separate video sequences and version the approved labels

Pipeline, step by step

Every prediction returns to the original frame before matching against ground truth.



Detection: precision and recall

First match predictions to eligible reference people, one-to-one, at a declared criterion.

Outcome	Meaning
True positive (TP)	A valid prediction matched to one reference person
False positive (FP)	An unmatched extra prediction, including a duplicate
False negative (FN)	A reference person with no valid matched prediction

Precision = TP / (TP + FP)

Illustration: TP = 8, FP = 2, FN = 4

8 / 10 = 80% correct predictions

Recall = TP / (TP + FN)

8 / 12 = 66.7% recovered targets

OKS: turn joint errors into a pose similarity

Object Keypoint Similarity: one score for a predicted pose and a reference person.

$OKS = (1 / N) \Sigma \exp[-d_i^2 / (2 A k_i^2)]$

Sum only over labelled reference joints ($v_i > 0$).

Symbol	How it enters the calculation
d_i	Euclidean distance from predicted joint i to its reference, in pixels
A	Ground-truth person area, in square pixels
$k_i = 2\sigma_i$	Fixed COCO joint-specific tolerance; not prediction confidence
N	Number of labelled reference joints contributing to this pose

Average Precision (AP) and Average Recall (AR)

COCO keypoints uses OKS, even where the evaluator prints the label “IoU”.



AP_t = mean interpolated precision at 101 recall levels

AP = mean AP_t for t = 0.50, 0.55, …, 0.95

AR = mean maximum recall over the same ten thresholds

AP50 / AP75 use one threshold; COCO retains up to 20 poses per image.

PCK: how many joints are close enough?

Percentage of Correct Keypoints; proposed scale: ground-truth person-box diagonal.

Normalized error: $e_i = \|p_i - g_i\|_2 / s$

PCK(α) = count(eligible joints with $e_i \le \alpha$) / N

N = all eligible GT joints; missing predictions count as incorrect.

Example: s = 200 px, α = 0.05 → tolerance = 10 px

Illustration: 8 eligible joints

6 within tolerance
PCK = 6 / 8 = 75%

7 confidently predicted
Coverage = 7 / 8 = 87.5%

Estimating the processing effect

Paired comparisons use the same reference population in every condition.

$$\Delta \text{ score} = \text{score(processed input)} - \text{score(baseline)}$$

1. **Fixed denominators**
   Count missed targets and unpredicted eligible reference joints
2. **Separate error sources**
   Report detection, pose localization and pitch-filter errors
3. **Independent confirmation**
   Choose settings on development clips, then test held-out sequences

Reproducibility safeguards

A result needs a recoverable input, configuration and execution environment.

Already recorded

Input and checkpoint SHA-256
Expanded YAML and processing order
Git revision and package versions
Hardware metadata and runner logs
Raw predictions and decision archives

Final-study verification

Pin actual container image digests
Freeze labels and evaluation settings
Repeat a reference run
Declare numerical tolerances
Measure cold and warm runtime separately

Experimental matrix

One unchanged baseline and eleven individual processing conditions

Processing family	Conditions
Scale and regions	Bicubic ×2, FSRCNN ×2, 2 × 2 tiling, detector-guided crops
Contrast and brightness	CLAHE, gamma darken, gamma brighten
Noise and restoration	Non-local means, bilateral denoising, unsharp, Wiener deblur

YOLO26x Pose
OpenPose BODY_25
HRNet-W32

Current football pilot
12 conditions × 3 models
782 identical source frames

Current experiment status

Exploratory detection yield on the football clip. Accuracy evaluation remains pending.

Model	Baseline on pitch	Example condition	On pitch
YOLO26	357	Bilateral denoising	2,115
OpenPose	11,647	CLAHE	15,338
HRNet	3,602	Bicubic ×2	15,554

Completed: 35 of 36 model jobs

Open cases: HRNet tiling, YOLO cropping and the failed OpenPose crop job

Next scientific milestones

1 **Reference-data agreement**

Confirm football joints, tracking timestamps and annotation fallback

2 **Protocol freeze**

Select the primary metric, matching rules and held-out sequences

3 **Verified ablation study**

Resolve runner issues, repeat baselines and score every condition

4 **Confirmation and downstream use**

Test selected combinations, then compare pitch positions with tracking

Backup: on-pitch screening overview 1/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Baseline	357 (reference)	11,647 (reference)	3,602 (reference)
Resize ×2	283 (-20.7%)	10,934 (-6.1%)	15,554 (+331.8%)
CLAHE	1,156 (+223.8%)	15,338 (+31.7%)	3,995 (+10.9%)
Gamma darken	315 (-11.8%)	12,193 (+4.7%)	3,634 (+0.9%)
Gamma brighten	440 (+23.2%)	11,029 (-5.3%)	3,568 (-0.9%)
NLM denoising	615 (+72.3%)	8,054 (-30.8%)	3,037 (-15.7%)

Backup: on-pitch screening overview 2/2

Record count and change relative to each model's own baseline

Processing	YOLO26	OpenPose	HRNet
Bilateral denoising	2,115 (+492.4%)	12,581 (+8.0%)	3,216 (-10.7%)
Unsharp	491 (+37.5%)	11,860 (+1.8%)	2,675 (-25.7%)
Motion deblur	1,627 (+355.7%)	7,362 (-36.8%)	214 (-94.1%)
Super-resolution ×2	312 (-12.6%)	11,583 (-0.5%)	15,565 (+332.1%)
Tiling	3,402 (+852.9%)	16,072 (+38.0%)	1 (-100.0%)
Detector crop	0 (-100.0%)	FAILED	3,467 (-3.7%)

Backup: PCP — Percentage of Correct Parts

Proposed strict-endpoint convention; other PCP variants must not be mixed with it.

Reference limb length: $L = \|g(a) - g(b)\|_2$

Correct limb if BOTH endpoint errors $\leq \alpha L$

PCP = correct eligible limbs / all eligible reference limbs

Illustration: $L = 40$ px and $\alpha = 0.5 \rightarrow 20$ px tolerance

Endpoint errors 12 px and 18 px: pass
Endpoint errors 12 px and 22 px: fail

Backup: PCK-AUC summarizes tolerances

AUC here is the area under PCK versus distance threshold, not the precision-recall curve.

$$\text{Normalized AUC} = (1 / \alpha_{max}) \times \int_{0}^{\alpha_{max}} \text{PCK}(\alpha)\, d\alpha$$

Threshold α	0.00	0.05	0.10
Illustrative PCK	0.00	0.50	1.00

Trapezoid estimate over [0, 0.10]

$$\text{AUC} \approx [0.05 \times (0 + 0.50)/2 + 0.05 \times (0.50 + 1)/2] / 0.10$$

$$= 0.50 \quad (50\% \text{ of the maximum possible area})$$

Backup: coverage and tracking-position error

Completeness and ground-plane localization answer different questions.

Joint coverage

$$\text{Coverage}(\tau) = \text{count}(\text{matched eligible joints with } c_i \geq \tau) / N$$

Use the same eligible GT joints; report the joint threshold and missing-target rule.

Player position in pitch coordinates

$$E_j = \sqrt{[(\hat{x}_j - x_j)^2 + (\hat{y}_j - y_j)^2]} \quad \text{metres}$$

Mean error = $\Sigma E_j / M$; also report median and matched-player coverage.

Backup: AP from a confidence-ranked list

Illustration at one OKS threshold: 3 reference people, 5 predictions, no ignored cases.

Rank / score	Outcome	Cum. TP / FP	Precision	Recall
1 / 0.95	TP	1 / 0	1.000	0.333
2 / 0.90	FP	1 / 1	0.500	0.333
3 / 0.80	TP	2 / 1	0.667	0.667
4 / 0.70	TP	3 / 1	0.750	1.000
5 / 0.60	FP	3 / 2	0.600	1.000

Precision envelope: 1.00 up to recall 1/3; then 0.75

$AP_t = (34 \times 1.00 + 67 \times 0.75) / 101 = 0.834$; $R_{max,t} = 1.00$

Backup: OKS scale and visibility example

Illustrative tolerances below are not the COCO sigma values for a particular named joint.

Joint error d	GT area A	Tolerance k	Contribution $\exp[-d^2/(2Ak^2)]$
0 px	10,000 px^2	0.10	1.000
10 px	10,000 px^2	0.10	0.607
20 px	10,000 px^2	0.10	0.135

For these 3 labelled joints: OKS = (1 + 0.607 + 0.135) / 3 = 0.581

GT visibility: 0 = unlabelled; 1 = labelled, occluded; 2 = labelled, visible.
Both 1 and 2 contribute. Predicted confidence is not a visibility label.

Backup: architecture-specific bottlenecks

All methods share an output task; their internal representations and failure modes differ.

Pipeline	Internal mechanism	Relevant limitation
HRNet top-down	Parallel high/low-resolution branches Repeated fusion → joint heatmaps	Missed detector boxes give no pose Pose work scales with crop count
OpenPose bottom-up	Joint maps + part-affinity vectors Group connected limbs into people	Tiny or occluded joint evidence Ambiguous grouping in crowds
YOLO26 single-stage	Shared image features Person boxes + keypoint predictions	Missed instances and localization Depend on scale and image features

Our comparison maps BODY_25 to the shared COCO17 joint schema.
A fixed input-size setting does not mean the three networks see identical representations.

Backup: freeze the calculation protocol

Official COCO scoring is implemented; supplementary football scoring remains to be finalized.

Decision	Explicit rule to freeze
Targets and matching	Same reference people; box-IoU detection versus pose-OKS matching
Joint eligibility	Label visibility, uncertainty and common COCO17 joint order
PCK / PCP	Scale, threshold, limb list, ≤ versus < and micro/macro averaging
PCK-AUC	Threshold interval, sampling grid and integration convention
End-to-end versus conditional	Missed people stay in end-to-end denominators; label matched-only scores
Coverage and tracking	Joint-confidence τ; synchronized identities; metric axes; missing matches